Programming Assignment

SC4001 Neural Networks and Deep Learning

Deadline: October 12, 2026

- This assignment is to be done **individually**. You may discuss the questions with others but your submission must be **your own unique work**.
- Complete **both parts A and B;** and submit your solutions to Part A and Part B separately: reports in .pdf files and codes in .zip files as follows:

  PartA_report_YourName.pdf, PartA_codes_YourName.zip

  PartB_report_YourName.pdf, PartB_codes_YourName.zip

  via **NTULearn** before the **deadline**.
- You will be using **3-way data split** method for parameter estimation and model evaluation. Use the **Adam optimizer** for training and **early stopping** to terminate training throughout the assignment. Train each model for a maximum of 20 epochs with a **patience** of 4 epochs and a **minimum improvement** of 1e-4.
- Codes need to be in python (.py) or python notebooks (.ipynb), tidied, properly labelled, and ready to run. Students are responsible for ensuring that their codes run from the beginning to end. A **README file** must include environments, dependencies, seeds, and commands to run the codes.
- The assessment is based on **the reports** submitted and **the code**s will be used only for verification purposes. The reports must be concise, include answers only to the questions asked, and less than **6 pages** each. 10 marks of the assignment are reserved for the presentation of the solutions.
- Late submissions will be penalized: **5% for each day up to three days**.
- **Generative AI Declaration** must be included at the beginning of your report. If any GenAI tools (e.g., ChatGPT, GitHub Copilot, Claude, Gemini) were used at any stage for code generation, debugging, writing, or interpretation, state which tool(s), for what purpose, and how the output was verified. Undisclosed use of GenAI will be treated as an academic integrity violation.
- You are requested to post general queries on the assignment on the **Discussion Board**. Please email any questions specific to you to **cx4042@e.ntu.edu.sg**.
- **TAs** for this assignment are Ms. Yige Yan, Mr. Ajith Santhisenan, Mr. Deepank Girish, and Ms. Tiara Natasha.

Part A: Regression Problem

In this part, you will develop **deep neural networks** for predicting Singapore HDB resale prices. A complete PyTorch baseline **baseline.py** that runs end-to-end, has been provided. We will use publicly available data on HDB flat prices in Singapore, obtained from data.gov.sg. The original dataset is combined with other datasets to include more informative features and they are given in the '**hdb_price_prediction.csv**' file. Open **housing_baseline_data.zip** and follow the **README** instructions to extract the source and data files.

Use the data gathered from 2017-2020 for training, in 2021 for validation and model selection, and in 2022 for final testing.

Feature	Type	Explanation
month	Categorical (Int)	Which month the resale transaction was performed.
year	Categorical (Int)	Which year the resale transaction was performed. Used for the data split, not as a model input.
town	Categorical (Str)	An area zoned for public housing. Generally, a town consists of a self-sufficient group of HDB flats with amenities and a town centre for that town. Each town has a population of around 50,000 - 250,000.
full_address	Categorical (Str)	Address of the flat. **Not used** for modelling as other metrics derived from it are used instead (dist_to_nearest_stn, dist_to_dhoby).
nearest_stn	Categorical (Str)	Closest MRT station to the flat. **Not used** for modelling as other metrics derived from it are used instead (degree_centrality, eigenvector_centrality).
dist_to_nearest_stn	Numeric	Distance from the flat to the nearest MRT station, in kilometres. Flats near stations tend to fetch higher prices.
dist_to_dhoby	Numeric	Distance from the flat to Dhoby Ghaut MRT station, in kilometres. Dhoby Ghaut is chosen as it is centrally located. Flats in the Central region are typically more costly.
degree_centrality	Numeric	A metric (computed for the MRT station closest to the flat) that represents the degree of the node, i.e., how many edges are connected to the node. (Rationale: flats near 'interchange' stations - stations with more than 1 MRT line - offer more transport options and thus have higher value. Stations in the central areas tend to have > 1 MRT line too).
eigenvector_centrality	Numeric	A more global metric than degree_centrality as it captures neighbourhood information. When the eigenvector centrality of a node is high, the nodes adjacent to it are likely to have high values too.
flat_model_type	Categorical (Str)	Type of flat. See this reference for more details. You're not expected to understand all flat types.
remaining_lease_years	Numeric	HDB flats are originally sold by HDB with a 99-year lease. Generally, with other variables held constant, flats with higher remaining lease will fetch a higher value. The original data was stored in years and months – this was turned into a scalar by converting it into months and dividing by 12.
floor_area_sqm	Numeric	Size of the flat in square meters. Bigger flats generally fetch higher prices.
storey_range	Categorical (Str)	Which floor the flat is at. For the same HDB block, flats on higher floors typically fetch higher prices.
resale_price	Numeric	Flat prices in Singapore Dollars. **Target to predict.**

Question A1 (13 marks)

Draw a block diagram of the baseline model and report the number of trainable parameters in it.

Perform hyperparameter optimisation on the baseline by using **grid search** with the **Ray** library. Optimise the hidden-layer width over {16, 32, 64, 128, 256} and the same embedding dimension over {2, 4, 8, 12, 16} for all four categorical features.

Select the configuration with the **lowest best validation RMSE** and report the optimal parameter values. For the selected configuration, show the plots of training and validation RMSE curves on the same axes.

Retrain the model selected on combined train and validation sets and evaluate the performance on the test set. Plot the training and testing RMSE curves on the same axes. Report final RMSE and R^2.

Using the evidences from the hyperparameter-search results and learning curves, discuss and justify the selected model configuration, including whether the model exhibits underfitting or overfitting.

Question A2 (13 marks)

Perform two ablation experiments below on the selected architecture from A1. Train the baseline and both ablations on the combined training and validation partitions of **five distinct seeds** of your choice. Use the same preprocessing, hyperparameters and number of epochs selected in A1 for all runs. Evaluate each trained model once on the test set and report the mean and standard deviation of test RMSE and R^2 across the five seeds.

a) Remove all four categorical embedding tables and use only the six continuous features as input. Adjust the first linear layer's input dimension accordingly and keep the hidden-layer width unchanged.

b) Retain both the categorical and continuous inputs, but replace the hidden ReLU activation with a Sigmoid activation.

Discuss and explain your conclusions from the ablation experiments, supporting with evidences from the results and plots of RMSE and R^2.

Question A3 (13 marks)

Evaluate the **wide-and-deep architecture** in Figure A against the baseline. Both branches receive the same concatenated embeddings and all the features. The linear branch has a single linear layer with one output and the MLP branch uses the selected architecture in Part A1. The final prediction is obtained by a linear sum of the outputs of the two branches:

$$\text{output} = \text{MLP output} + \lambda \times \text{linear output}$$

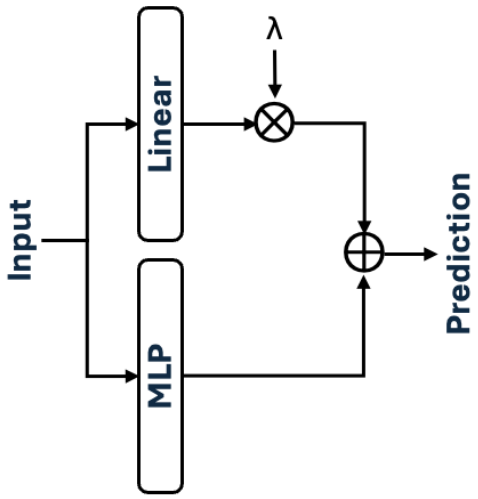


Figure A. Wide and deep architecture

Find the optimal value of parameter λ in {0, 0.25, 0.5, 1, 2}, keeping the selected A1 hyperparameters fixed.

Evaluate the selected wide-and-deep configuration on five different seeds. Report the mean and sample standard deviation of test RMSE and R^2, with relevant plots, and compare against the A1 baseline results.

Explain the *mechanism* by which the change could affect optimisation or generalisation, and determine whether the evidence is consistent with that mechanism. Discuss at least one *limitation* or confound in the comparison.

Question A4 (6 marks)

Use Captum's **FeatureAblation** method on the test set to explain the final baseline model from A1 and the wide-and-deep model from A3. Use the same seed to train both models.

- For each model, compute the FeatureAblation attribution of each of the ten original input features for every test sample. Then compute the global feature importance score for each feature as the mean absolute attribution across all test samples.
- Present the ten resulting feature importance scores in Singapore dollars in a labelled bar plot, ordered from highest to lowest.

Report the three most important features for each model. Discuss and compare the results, and identify one limitation of the analysis.

Part B: Classification Problem

In this part, we try to classify chest X-ray images from healthy subjects and pneumonia patients. The dataset is provided in **chestxray_binary_64.npz**. The dataset is already partitioned into training, validation, and test partitions.

The grey-level image intensities are to be scaled to [0, 1] and then normalized using the mean and standard deviation evaluated on the training set.

Parts B1–B3 involve building the CNN architectures shown in Figure B. Use **batch normalization** and **ReLU** activation after each convolutional layer. Use the bias in the fully connected layers, but **do not use** bias in convolutional layers.

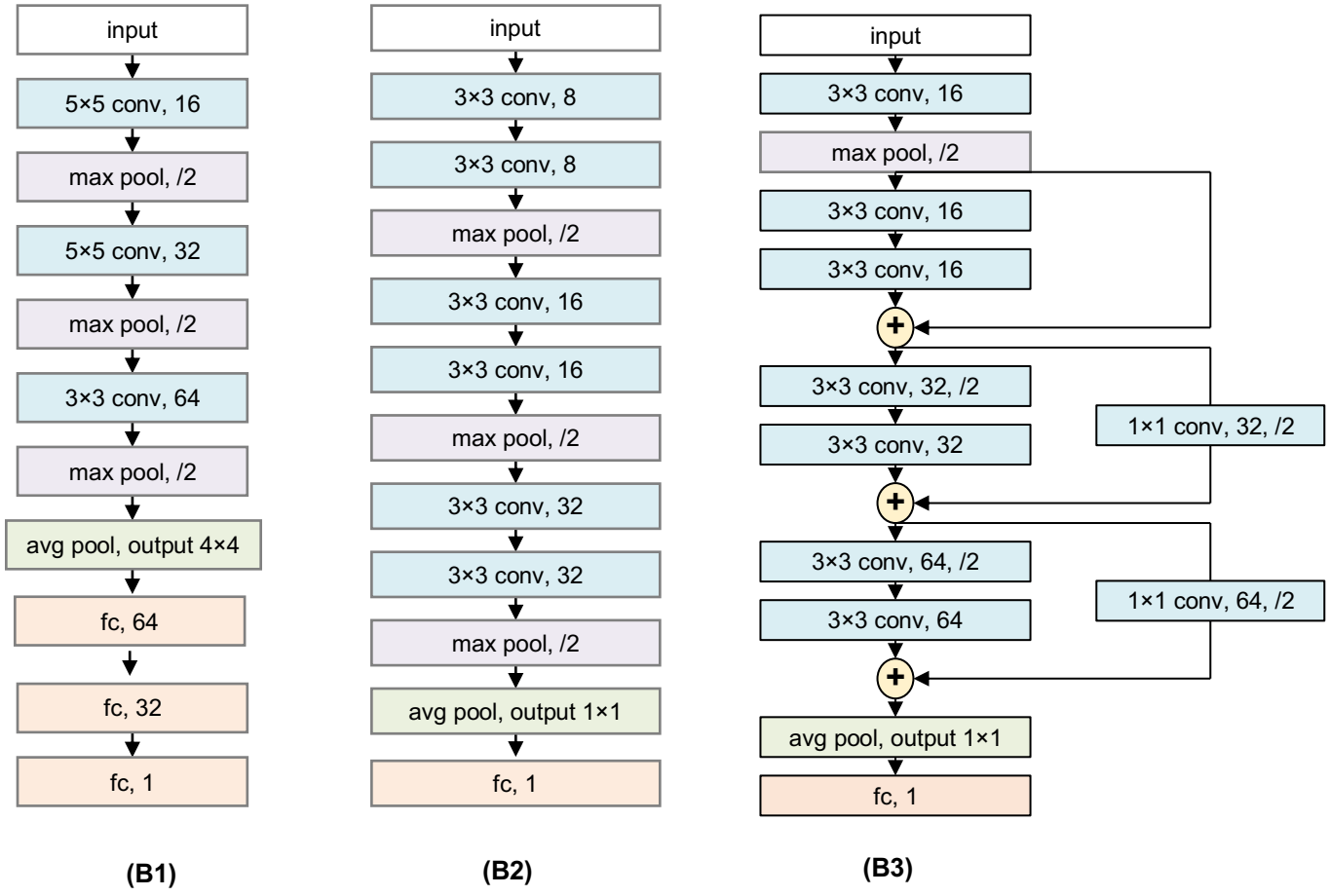


Figure B. (B1) AlexNet like architecture, (B2) VGG like architecture, and (B3). ResNet like architecture. All the layers use ReLU activation function including after residual addition ⊕. All convolutional layers use BatchNorm and SAME padding. Layers indicating /2 have a stride of two and all other layers have stride of 1. Dropout = 0.5 is applied only in the hidden fully-connected (fc) layers in B1 architecture.

Question B1 (13 marks)

Implement and train an **AlexNet-like architecture** shown in Figure B1. Use a batch size of 128 and train the network with learning rates from {1e-5, 1e-4, 1e-3, 1e-2, 1e-1}.

Plot validation (i) loss and (ii) accuracy against epochs for the five learning rates in two separate figures.

Present a table containing the learning rate, best epoch, validation loss, validation classification error (1 − accuracy), sensitivity, and specificity.

Select the optimal **learning rate** based on the lowest validation classification error. Support your decision using the reported results, explain what the curves show about convergence, and discuss why sensitivity and specificity are also needed to interpret model performance.

Evaluate the final model once on the test set and report its accuracy, sensitivity, and specificity.

Question B2 (13 marks)

In this question, you will determine the optimal **batch size** for a **VGG-like architecture** shown in Figure B2. Use the best learning rate found in **Part B1** and train the network with batch sizes from {16, 32, 64, 128, 256}.

Plot validation (i) loss and (ii) accuracy against epochs for the five batch sizes in two separate figures.

Present a table containing the batch size, best epoch, validation loss, validation classification error (1 − accuracy), sensitivity, and specificity.

Select the optimal **batch size** based on the lowest validation classification error. Explain the reasons for your selection using the reported results.

Evaluate the final model once on the test set and report its accuracy, sensitivity, and specificity.

Question B3 (13 marks)

In this question, you will build for a compact **ResNet-like architecture** shown in Figure B3. Use the best learning rate obtained in part B1 and batch size in part B2 to train the network.

Report the accuracy, sensitivity, and specificity of the model.

Compare the architectures developed in B1-B3 in a table containing accuracies, sensitivities, specificities, classification error, number of learnable parameters, and floating-point operations. State the pros and cons of different architectures.

Question B4 (6 marks)

In pneumonia cases, chest X-rays show abnormal white and cloudy patches. In this question, we explore if XAI methods can identify these patches by using **Grad-CAM** technique. Grad-GAM is available in **Captum** package in Pytorch.

Select randomly FIVE representative X-rays from pneumonia patients. Implement Grad-CAM on architectures developed in B1-B3.

Overlay Grad-CAM heatmaps on X-ray images and compare the results.